SUB-GHZ / LONG-RANGE RADIOS

MODULE	BAND	RANGE	RATE	POWER	COST	VERDICT
SX1262 (E22-900M30S)	915 MHz	5–15 km	0.3–50 kbps	22 dBm	$15–25	MESH STANDARD — fleet default
SX1276	868/915 MHz	similar	0.3–50 kbps	20 dBm	$10–18	alt — older, more libs
CC1101	315/433/868/915	1–3 km	1.2–500 kbps	10 dBm	$5–10	sensor motes — short range
nRF905	433/868/915	2–5 km	20–100 kbps	13 dBm	$8–15	legacy — superseded by SX1262
LoRa32 / RFM95	915 MHz	same radio	same	14–20 dBm	$20–35	dev boards only

CELLULAR / BACKHAUL

MODULE	GEN	RATE	GNSS	COST	POWER	VERDICT
SIM7600	LTE Cat-4	150 Mbps	yes	$60–90	~1 W avg	L1/TOTEM/ANCHOR backhaul
SIM7070	LTE Cat-1	10 Mbps	yes	$40–60	~0.5 W	MITE/PHONE low-power
BG96	LTE Cat-M1/NB-IoT	1 Mbps	no	$25–40	~0.1 W	sensor backhaul — tiny data
Starlink Mini	LEO sat	100+ Mbps	—	$300–600	~30 W	ANCHOR true dead zones
Iridium Certus	L-band	22–128 kbps	—	OEM $500+	~5 W	PHONE SOS + north

DECISION RULES

— Mesh = SX1262 everywhere — one radio, one firmware path, fleet commonality.

— Backhaul = LTE where towers exist (SIM7600), NB-IoT where they barely do (BG96). SOS = Iridium-class L-band — the only global guarantee.

— Starlink = Anchor-level only — power-hungry, business case required.

— No module gets chosen without: Canadian availability, -30 °C rating, supply chain ≥2 sources.

ANTENNA OPTIONS

TYPE	GAIN	USE
¼-wave whip SMA	2–5 dBi	L1/Anchor field replaceable
PCB meander	1–2 dBi	Mite — space-constrained
Helical	3–8 dBi	Relay-bee high perch
Yagi directional	8–12 dBi	Anchor backbone backhaul
LDS (laser-direct)	—	Phone — no external whips

REJECTED OPTIONS (AND WHY)

— Wi-Fi HaLow — promising but ecosystem immature; revisit rev-C.

— Meshtastic commodity modules — closed firmware, no signed envelopes.

— Cellular-only 'solution' — fails exactly where ArcticChain lives.

— Satellite broadband for nodes — watts and dollars for kbps of need.

— Proprietary mesh radios — vendor lock-in kills community ownership.

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SCALE **NTS**
SHEET **1/2**
DWG NO **AF-X-101**
UNITS **mm**

COMPUTE / MCU

CHIP	CLASS	PRICE	POWER	USE	VERDICT
ESP32-S3	MCU+WiFi	$3–8	~0.5 W	Mite/swarm	DEFAULT for nodes
nRF52840	MCU+BLE	$4–8	~0.1 W	ultra-low-power	seed motes candidate
STM32WB	MCU+BLE	$5–10	~0.2 W	industrial	alt for certified builds
CM5	SBC Linux	$60–90	~5 W	L1/Totem/Anchor	gateway compute
RK3326	SBC Android	$40–70	~4 W	phone-class	Phone option B

POWER STORAGE

CHEMISTRY	ENERGY	CYCLE LIFE	COLD RATING	VERDICT
18650 Li-ion	3 Ah/cell	300–500	-20 °C derate	L1 portable
LiFePO4	3.2 V/cell	2000+	-30 °C ok	STATIONARY default
LiPo pouch	varies	300–500	-20 °C	Mite/Phone compact
Printed battery	µWh	season	—	Seed motes (R&D)

DISPLAY

TYPE	POWER	SUNLIGHT	USE
E-Ink 7.5–13.3in	zero idle	perfect	Totem/L1 public
Transflective LCD	low	good	Phone outdoor
OLED	med	poor	consumer look — wrong for field
LED matrix	low	ok	status only — not content

ENCLOSURE / MECHANICAL

OPTION	COST	TOOLING	VERDICT
FDM print PETG/ASA	low	none	prototypes + field spares
Injection mold PC/ABS	high NRE	mold $5–20k	volume products
Polycarbonate box	med	none	IP66 infrastructure
Sheet metal + gasket	med	fab	Anchor industrial
3D-print + TPU overmold	low-med	print	Phone rev-B option

CONNECTORS / INTERFACES

STANDARD	USE	WHY
USB-C PD	power + data	universal, glove-friendly
Pogo pins	dock/module	blind-mate, no doors
SMA/U.FL	antennas	field-replaceable
JST-PH	battery	polarized, tool-light
M2/M3 screws	enclosure	hand-tool service
Magnetic catch	lids	tool-less battery access

THE PLATFORM RULE

— Common compute (ESP32-S3/CM5), SX1262 mesh, USB-C, signed envelope — everywhere.

— Diversity only where environment demands: battery, display, enclosure.

— Every choice must pass: Canadian supplier? -30 °C? hand-tool repair? community-open?

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